NurseWatch-AI: Agentic Shift Safety and Early Warning System

1. Core Problem

In hospitals, patients often deteriorate gradually. A single BP, sugar, heart rate, SpO_2, or respiratory-rate value may not look dramatic, but the **trend across shifts** may show a clear worsening pattern. The risk is highest during shift changes because information may be fragmented, missed, or poorly handed over.

So the goal should not be only a dashboard. The system should act like an **agentic clinical safety workflow** that continuously asks:

> "Is this patient becoming worse across shifts, and has the right person acknowledged and acted on it?"

This aligns with patient-safety thinking. WHO's Global Patient Safety Action Plan 2021–2030 focuses on reducing avoidable harm in healthcare, and early-warning systems are designed to detect deterioration and trigger timely response.

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2. System Objective

To build an **agentic AI and autonomous workflow system** that monitors patient vitals across nursing shifts, detects absolute danger values and hidden worsening trends, generates structured handover summaries, escalates unacknowledged alerts, and maintains an audit trail of all nurse and clinician actions.

The system should support decision-making but not replace clinical judgment.

Clinical disclaimer:
For nursing decision support only. All abnormal trends must be verified by a nurse/doctor. The AI should not independently diagnose, prescribe, or change treatment.

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3. Why Agentic AI Is Needed

A normal dashboard only shows information. An **agentic system** takes responsibility for workflow completion.

Normal dashboard	Agentic AI workflow
Shows vitals	Watches vitals continuously
Displays red values	Detects trends across shifts
Depends on nurse noticing	Pushes alert to responsible nurse
No follow-up	Checks if alert was acknowledged
No escalation	Escalates if no action is recorded
Manual handover	Auto-generates SBAR handover
Poor accountability	Creates audit log
Single-patient view	Ward-level risk prioritization

NHS guidance on acute deterioration emphasizes clear escalation processes: who to escalate to, expected response, expected timeframes, and mechanisms for re-escalation or de-escalation as the patient worsens or improves.

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4. Overall Architecture

A. Data Sources

Data source	Description
Bedside monitor	HR, SpO_2, respiratory rate, temperature
Manual nursing entries	BP, glucose, pain score, urine output, GCS if added
HIS / EMR	diagnosis, ward, bed, doctor, nursing shift, medication orders
Lab system	Hb, WBC, creatinine, electrolytes, lactate if available
Medication system	insulin, antihypertensives, antibiotics, oxygen therapy
Nurse action log	rechecked vitals, informed doctor, medication given, patient shifted

Mock data generator	initial prototype: 50 patients × 14 days × 3 shifts

B. Core Modules

Module	Function
Vital ingestion agent	Collects vitals from devices or manual entries
Patient context agent	Pulls diagnosis, age, risk category, ward, doctor
Rule engine agent	Applies absolute threshold, trend, persistent abnormality rules
Early-warning score agent	Calculates NEWS2-like score or hospital-approved EWS
Trend reasoning agent	Detects worsening across shifts
Handover agent	Generates SBAR summary
Escalation agent	Sends alert to nurse, supervisor, duty doctor
Closure agent	Checks whether action was taken
Audit agent	Maintains legal and quality log
Quality analytics agent	Produces ward-level safety reports

NEWS2 was developed to improve detection and response to clinical deterioration in acutely ill patients, and such standard early-warning approaches can be adapted locally after clinical approval.

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5. Vital Parameters to Track

Basic prototype vitals

Parameter	Example danger rule
Systolic BP	>180 or <90
Diastolic BP	>110 or very low with symptoms
Blood glucose	>300 or <70
Heart rate	>120 or <50
Temperature	>38.5°C or <35°C
SpO_2	<92%, or lower threshold for COPD if doctor-approved
Respiratory rate	>24 or <8
Pain score	≥7 or sudden increase
Consciousness	confusion, drowsiness, reduced response
Urine output	low output over shift, if available

Extended hospital version

Add:

- oxygen requirement;
- GCS / AVPU;
- urine output;
- fall risk;
- sepsis risk;
- post-operative bleeding signs;
- lactate, WBC, creatinine;
- insulin administration and hypoglycemia events;
- vasopressor/oxygen escalation where applicable.

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6. Alarm Logic

A. Absolute threshold alerts

These trigger when a value crosses a dangerous limit.

Alert	Rule	Severity
Severe hypertension	SBP >180	Red
Hypotension	SBP <90	Red
Hyperglycemia	Glucose >300	Red
Hypoglycemia	Glucose <70	Red
Tachycardia	HR >120	Yellow/Red depending context
Bradycardia	HR <50	Yellow/Red depending context
Low oxygen	SpO_2 <92	Red
High respiratory rate	RR >24	Yellow/Red
High fever	Temp >38.5°C	Yellow/Red

B. Trend alerts

These detect deterioration even before an emergency threshold is crossed.

Trend	Rule	Example
BP worsening	Two or more consecutive shifts with >10% SBP rise/drop	130 → 146 → 164
Sugar worsening	Two or more consecutive shifts with >20 mg/dL rise	160 → 190 → 225
HR worsening	Persistent increase over 3 shifts	88 → 105 → 118
SpO_2 worsening	Fall across shifts	97 → 94 → 92

Fever trend	Increasing temperature	37.5 → 38.1 → 38.7
Pain trend	Pain increasing despite medication	3 → 6 → 8

C. Persistent abnormality alerts

These catch patients who are “not critical yet” but remain unsafe.

Persistent condition	Rule
BP persistently high	Three shifts above 140/90
Glucose persistently high	Three shifts above 180
Fever persists	Fever across 2–3 shifts
SpO_2 persistently low	Repeated values below ward threshold
Pain uncontrolled	Pain ≥6 for two shifts

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7. Agentic Workflow

Step 1: Vital capture

The **Vital Ingestion Agent** receives or asks for vital entries at fixed times:

- Morning shift
- Evening shift
- Night shift

If vitals are missing, the system sends a reminder to the assigned nurse.

Example:
“Vitals for Bed 1204, Patient A, night shift not entered. Please update or mark reason.”

Step 2: Data validation

The **Validation Agent** checks for impossible or suspicious values.

Examples:

- BP 800/60 → likely entry error.
- SpO_2 40 in stable patient → recheck urgently.
- Glucose 35 → urgent recheck and hypoglycemia protocol.
- Same exact vitals repeated across many shifts → possible copy-forward issue.

Step 3: Rule-based detection

The **Clinical Rule Agent** applies:

1. absolute threshold rules;
2. trend rules;
3. persistent abnormality rules;
4. patient-specific risk rules.

Example:

> Patient with diabetes + post-op status + glucose 260 for 3 shifts = orange alert, even if not >300.

Step 4: Early-warning scoring

The **EWS Agent** calculates a hospital-approved early warning score using respiratory rate, oxygen saturation, temperature, systolic BP, heart rate, consciousness, and oxygen support.

This should be approved by Amrita clinicians. It can be based on NEWS2 principles but localized to hospital policy.

Step 5: Plain-English summary

The **LLM Summary Agent** generates a short handover note.

Example:

> "Patient's systolic BP increased from 142 to 158 to 176 over three shifts. Glucose remained above 200 for the last three readings. No nurse action recorded after the evening alert. Clinical review recommended."

The LLM should only summarize retrieved structured data. It should not invent diagnosis or treatment.

Step 6: Escalation

The **Escalation Agent** sends alert to the correct role.

Severity	Recipient	Expected response
Green	Dashboard only	Routine monitoring
Yellow	Assigned nurse	Acknowledge and recheck
Orange	Nurse + nursing supervisor + duty doctor	Review within defined time
Red	Nurse + duty doctor + rapid response / ICU team	Immediate action

The system should avoid alarm fatigue. The Joint Commission has highlighted medical-device alarm safety as a major hospital safety issue; therefore alerts must be prioritized, actionable, and auditable, not just noisy. (Joint Commission)

Step 7: Closure tracking

This is where the system becomes truly agentic.

The **Closure Agent** checks whether action was taken.

Possible nurse actions:

- rechecked vitals;
- informed duty doctor;
- doctor reviewed;
- medicine given as per order;
- oxygen started as per order;

- sample sent;
- patient shifted;
- false alert / measurement error;
- patient refused;
- other note.

If no action is recorded within the allowed time, escalation continues.

Example:

> “Orange alert for Bed 1204 not acknowledged for 15 minutes. Escalating to nursing supervisor.”

Step 8: Shift handover

At shift change, the **Handover Agent** generates ward-wise and patient-wise SBAR summaries.

SBAR structure

Section	Content
Situation	Current alert and patient condition
Background	Diagnosis, age, risk factors
Assessment	Vitals trend and risk level
Recommendation	Recheck, doctor review, monitor closely, urgent escalation

Example:

Situation: Bed 1204 has orange alert due to rising BP and persistent hyperglycemia.
Background: 62-year-old diabetic patient, post-operative day 2.
Assessment: SBP 148 → 164 → 178; glucose 210 → 236 → 262 over three shifts.
Recommendation: Recheck vitals, inform duty doctor, monitor glucose and BP as per protocol.

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8. Autonomous Agents in the System

Agent	Responsibility
Shift Watch Agent	Monitors each patient across shifts
Missing Vitals Agent	Finds missing or delayed vitals
Trend Detection Agent	Detects deterioration patterns
Risk Scoring Agent	Assigns green/yellow/orange/red
Handover Agent	Creates SBAR notes
Escalation Agent	Notifies responsible staff
Closure Agent	Checks alert acknowledgement and action
Audit Agent	Records every alert, action, delay, and escalation
Quality Agent	Creates ward-level safety reports
Learning Agent	Reviews false positives and proposes rule tuning, after clinician approval

The **Learning Agent** should not automatically change clinical rules. It can only recommend changes for quality committee review.

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9. Dashboard Design

A. Ward-level view

Bed	Patient	Risk	Main issue	Last action	Time pending
1204	Patient A	Red	SBP 190, chest pain	Doctor informed	3 min
1211	Patient B	Orange	Glucose rising 3 shifts	No action	18 min
1218	Patient C	Yellow	HR rising	Recheck done	Closed

B. Patient-level view

Show:

- 14-day trend chart;
- shift-wise vitals;
- alert history;
- nurse action log;
- AI summary;
- doctor notes;
- pending tasks.

C. Command-centre view

Show:

- total red alerts;
- unacknowledged alerts;
- ward-wise risk heatmap;
- average response time;
- repeated deterioration cases;
- shift handover risk list.

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10. Example Full Workflow

Scenario

Patient in Bed 1204 has diabetes and hypertension.

Shift	SBP	DBP	Glucose	HR	SpO_2
Morning	142	88	178	86	97
Evening	158	94	205	98	96
Night	176	100	238	110	95

AI detection

- BP rising >10% across shifts.
- Glucose rising >20 mg/dL across shifts.
- BP above 140/90 for three shifts.
- Glucose above 180 for two shifts.
- HR rising.

Alert

Orange alert: Patient worsening across shifts.

AI handover summary

> "Patient shows worsening cardiometabolic trend over three shifts: SBP increased from 142 to 176, glucose increased from 178 to 238, and HR increased from 86 to 110. BP has remained above 140/90 for three shifts. Please recheck vitals, verify medication adherence, and inform duty doctor as per ward protocol."

Agentic follow-up

- Alert sent to assigned nurse.
- Nurse must acknowledge.
- If not acknowledged in 15 minutes, nursing supervisor is alerted.
- If no action in 30 minutes, duty doctor is alerted.

- All events are logged.

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11. Safety Rules

The system must follow these principles:

1. **AI does not prescribe.**
2. **AI does not diagnose.**
3. **AI only recommends escalation, recheck, or clinical review.**
4. **Every alert must be explainable.**
5. **Every alert must be auditable.**
6. **Nurses and doctors can mark false positives with reason.**
7. **No silent closure of red alerts.**
8. **Thresholds must be approved by hospital clinicians.**
9. **Patient-specific thresholds must be possible.**
10. **Alarm fatigue must be monitored.**

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12. Summary

NurseWatch-AI is an agentic early-warning and shift-handover system for Amrita Hospital Faridabad. It continuously monitors vital-sign trends across nursing shifts, detects hidden deterioration, generates SBAR handover summaries, escalates unacknowledged alerts, and creates a complete audit trail for patient safety and quality improvement.